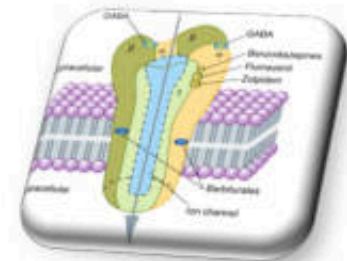
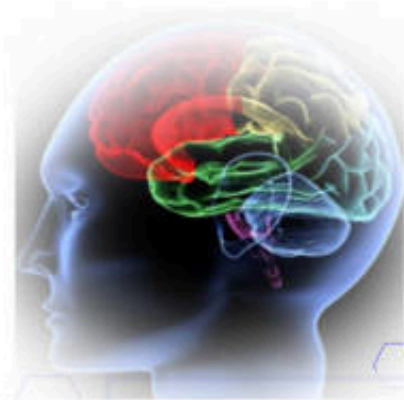


Hypnotics and Anxiolytics



Pongsatorn Meesawatsom
Department of Pharmacology
Faculty of Pharmacy
Mahidol University

วัตถุประสงค์เชิงพฤติกรรม

- * อธิบายกลไกการออกฤทธิ์และเปรียบเทียบผลทางเภสัชวิทยาของ **anxiolytics** ได้
- * อธิบายกลไกการออกฤทธิ์ เปรียบเทียบผลทางเภสัชวิทยาต่อวงจรการนอนหลับของ **hypnotics** ได้
- * ระบุน้ำหนักที่สำคัญของการให้คำแนะนำการใช้ **hypnotics** แก่ผู้ป่วยโดยใช้พื้นฐานทางเภสัชวิทยาได้

Disorders associated with fear and worry

Amygdala

Fear phobia, panic

Anxiety disorders

- Generalized anxiety disorder (GAD)
- Social anxiety disorder (SAD) or social phobia
- Specific phobia
- Panic disorder (PD)
- Separation anxiety disorder
- Selective mutism

Cortico-striato-thalamo-cortical (CSTC) pathways

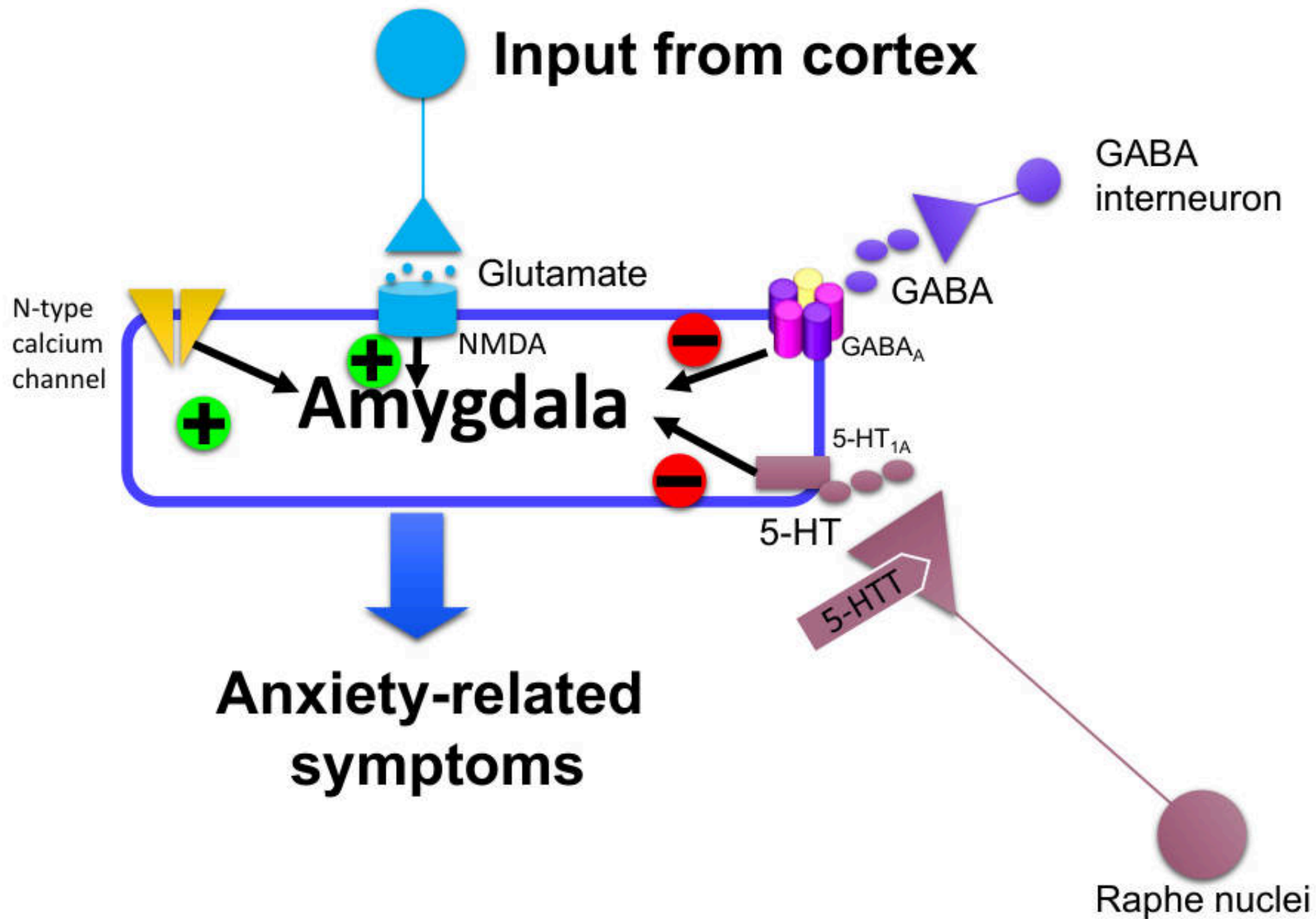
Worry Anxious misery
 Apprehensive expectation
 Obsession

Trauma- and stress-related disorders

- Post-traumatic stress disorder (PTSD), acute stress disorder

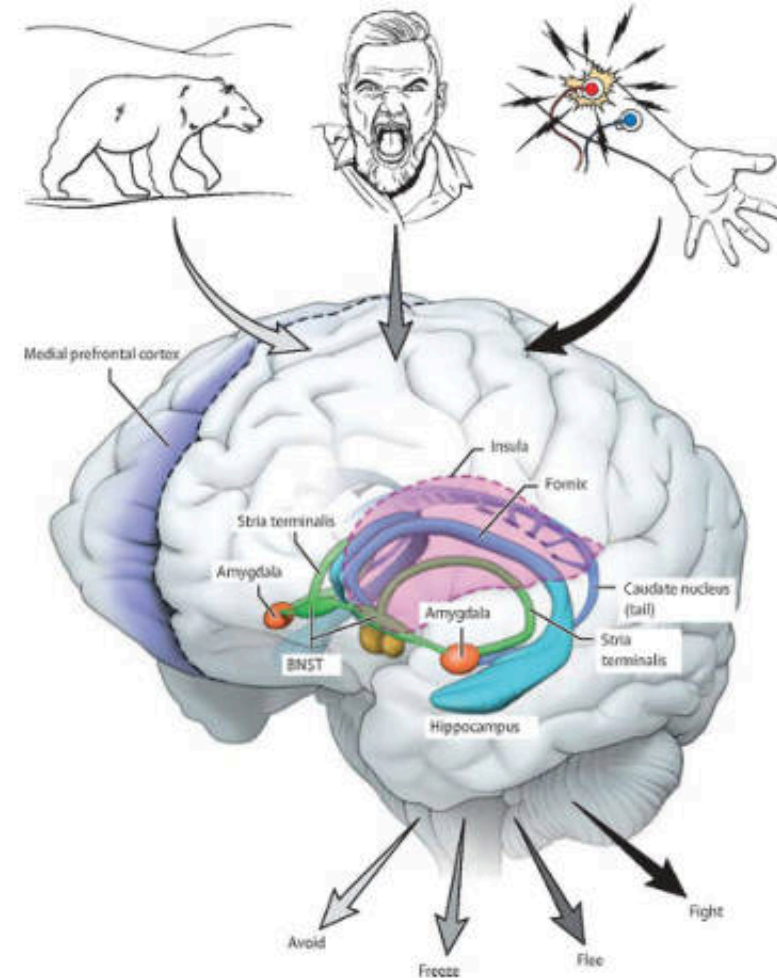
Obsessive-compulsive and related disorders

- OCD, trichotillomania, hoarding disorder



Anxiety-related symptoms

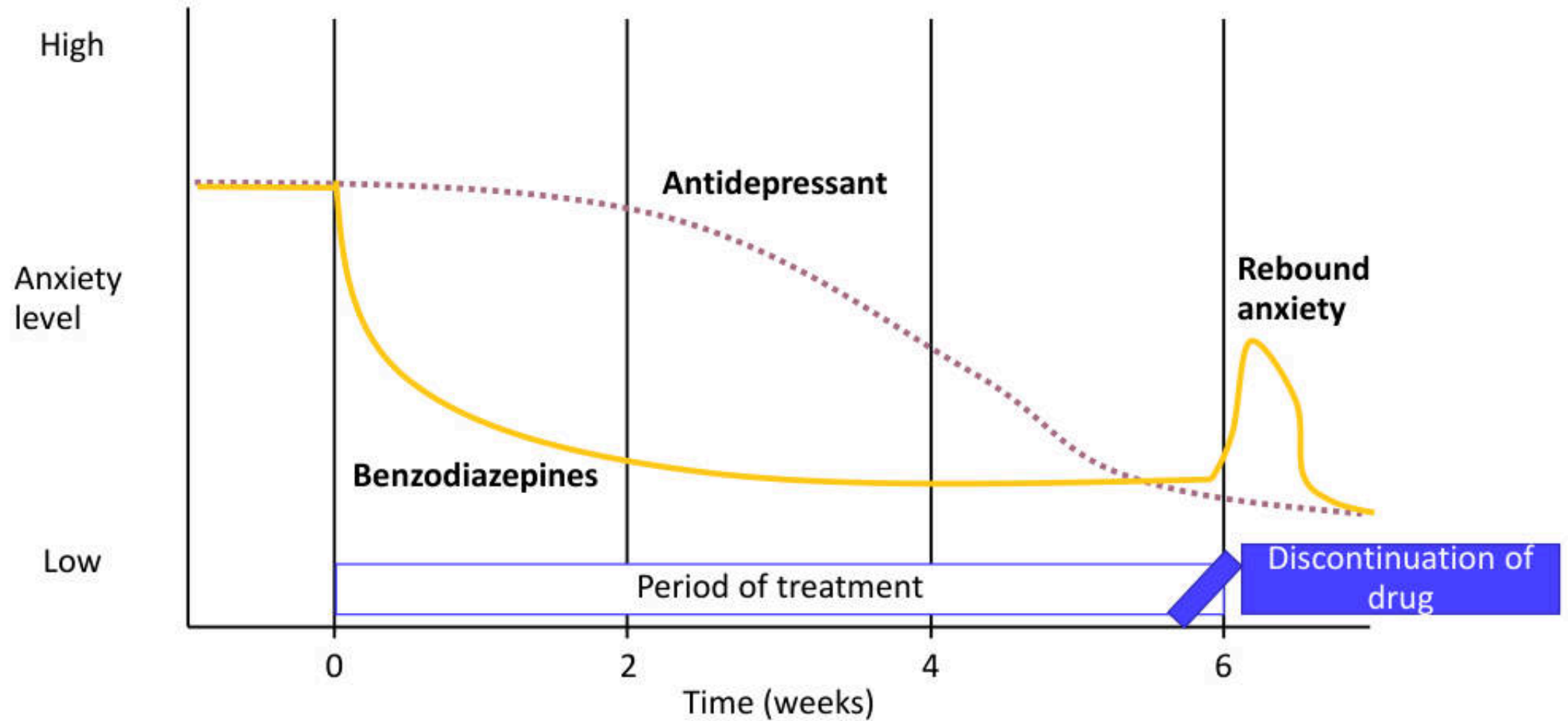
- ✧ Anxious mood, fear
- ✧ Autonomic symptoms
 - ★ Abdominal cramp, stomachache, peptic ulcer, nausea, vomiting, urination, defecation, bradycardia
 - ★ Palpitation, tremor, sweating, BP elevation, pupil dilation
- ✧ Other somatic symptoms
 - ★ Medical unexplained chest pain
 - ★ Freezing, facial expression of fear



Drugs for anxiety disorders

1. Antidepressants (onset 1-2 weeks) – increase serotonin
 - ★ Tricyclic antidepressants (TCA)
 - ★ Selective serotonin reuptake inhibitors (SSRIs)
 - ★ Serotonin-norepinephrine reuptake inhibitors (SNRIs)
2. Benzodiazepines (quick onset) – GABA_A APL
 - ★ PTSD and OCD are less responsive to BZPs
3. Pregabalin (Lyrica[®]) – Blocks N-type calcium channel
4. Etifoxine (Stresam[®])
 - ★ Psychosomatic manifestations of anxiety such as autonomic nervous system imbalance
5. 5-HT_{1A} agonist/partial agonist
 - ★ Buspirone (onset 1-2 weeks)
 - ◇ 5-HT_{1A} receptor partial agonist
6. Others
 - ★ β -blockers; propranolol

Time course of therapeutic effect of benzodiazepines, buspirone and antidepressants in anxiety disorders



Medication class		Mechanism of action	FDA approvals for anxiety disorder	Off-label uses	Therapeutic dose ranges (mg/day)
SSRIs:	Fluoxetine	Selective 5-HT reuptake inhibitor	PD	GAD, SAD	20–60
	Sertraline		PD, SAD	GAD	50–200
	Citalopram		None	GAD, PD, SAD	20–40
	Escitalopram		GAD	PD, SAD	10–20
	Paroxetine		PD, SAD, GAD	None	20–60
	Paroxetine ER		PD, SAD	GAD	27–75
	Fluvoxamine		None	GAD, PD, SAD	100–300
SNRIs:	Duloxetine	5-HT, NE (and DA) reuptake inhibitor	GAD	PD, SAD	30–60
	Venlafaxine (XR)		GAD	PD, SAD	75–300
	Desvenlafaxine		None	GAD, PD, SAD	50–100
TCAs:	Clomipramine	NE and 5-HT reuptake inhibitor	None	GAD, PD, SAD	100–250
	Imipramine		None	GAD, PD, SAD	100–300
	Desipramine		None	GAD, PD, SAD	100–200
	Nortriptyline		None	GAD, PD, SAD	50–150
MAOIs:	Phenelzine	MAO inhibitor	None	GAD, PD, SAD	30–90
Mixed antidepressants:	Mirtazapine	5-HT ₂ , 5-HT ₃ , α_2 , H ₁ antagonist	None	Anxiety, GAD, PD, SAD	15–45
GABAergic drugs:	Pregabalin	Unclear, may modulate Ca ²⁺ channels	None	GAD, SAD	150–600
	Gabapentin		None	GAD, SAD, PD	600–2,400
Benzodiazepines:	Clonazepam	GABA-A agonist	PD	Anxiety, GAD, PD, SAD	1–2
	Alprazolam		Anxiety, PD	GAD, PD, SAD	1–4
	Lorazepam		Anxiety	GAD, PD, SAD	2–6
	Chlordiazepoxide		Anxiety	GAD, PD, SAD	20–100
	Oxazepam		Anxiety	GAD, PD, SAD	30–60
Antipsychotics:	Trifluoperazine	D ₂ antagonist	Anxiety	GAD, PD, SAD	2–6
	Olanzapine	D ₂ , 5-HT ₂ H ₁ antagonist	None	Anxiety, GAD	5–15
	Quetiapine	D ₂ , 5-HT ₂ H ₁ antagonist	None	Anxiety, GAD	50–300
Beta-blockers:	Propranolol	β -1, β -2 antagonist	None	Anxiety, PD, SAD	60–120
Antihistamines:	Hydroxyzine	H ₁ antagonist	Anxiety	GAD, PD, SAD	25–100
Other anxiolytics:	Buspirone	5-HT _{1A} partial agonist	Anxiety	GAD	15–60

Insomnia

✧ Classification

★ By causes

- ◇ Primary insomnia
- ◇ Secondary insomnia

★ By onset

- ◇ Early (sleep onset)
- ◇ Late insomnia (sleep maintenance)

★ By time

- ◇ Transient or short-term
- ◇ Chronic insomnia

Causes of insomnia

Psychological

Increased arousal due to:

- stress or worry
- bereavement
- being 'on call'
- noise
- children

Pharmacological

Prescribed drugs, e.g.:

- β -blockers
- antidepressants
- steroids

Non-prescribed drugs, e.g.:

- alcohol
- caffeine
- stimulants
- withdrawal

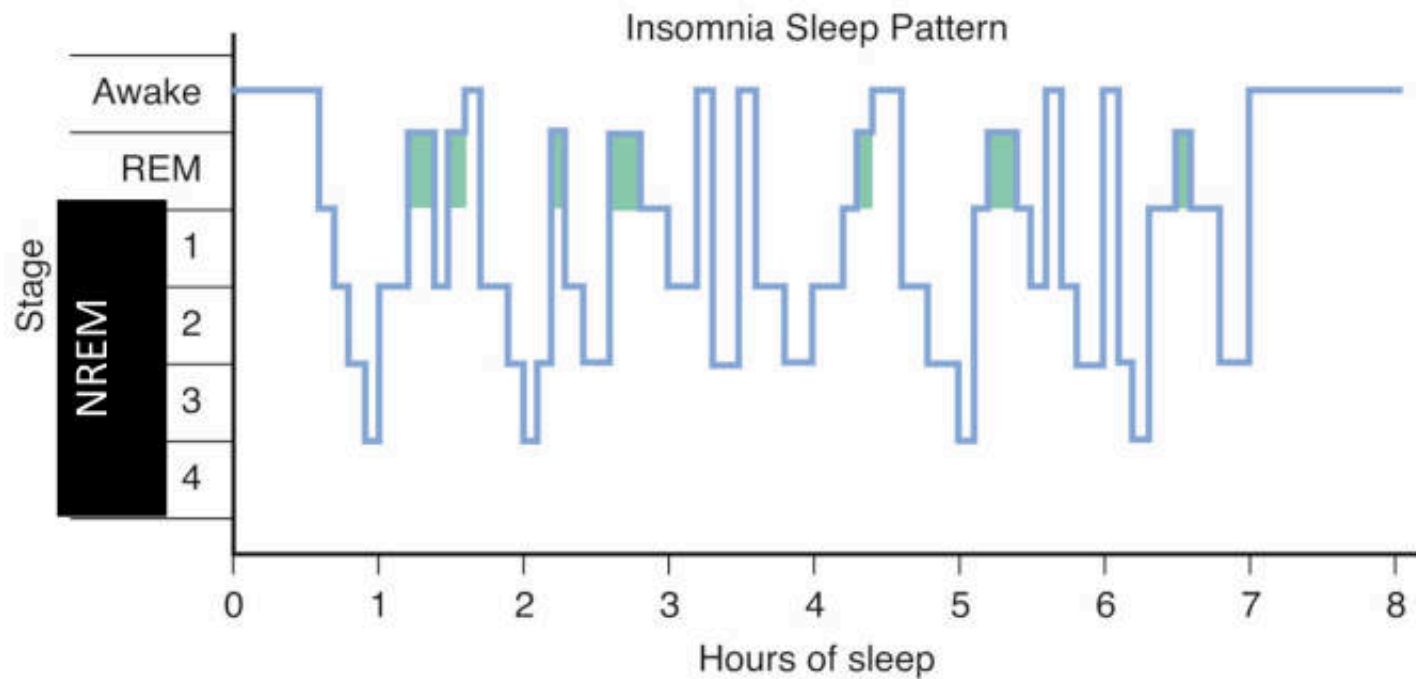
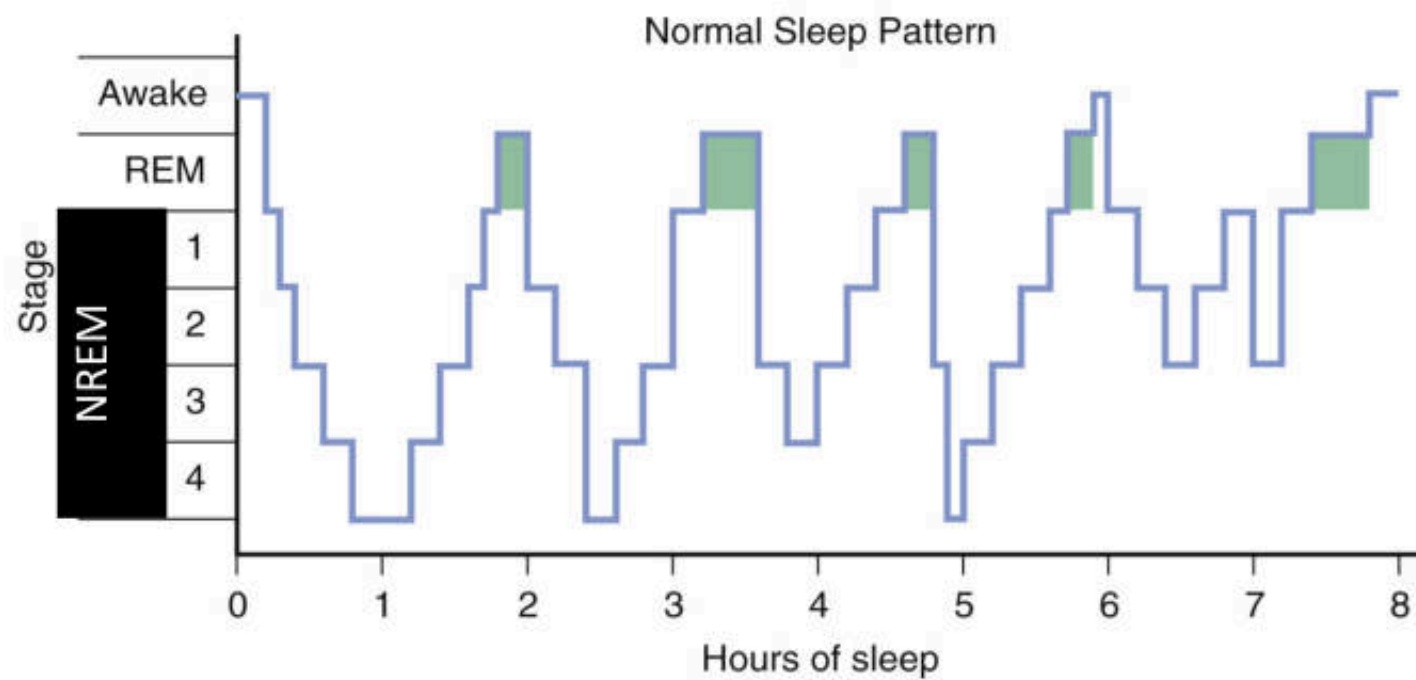
Physical

- Pain
- Pregnancy
- Illness, especially cardiorespiratory
- Urinary problems

- Obstructive sleep apnea (OSA)

Psychiatric

- Depression
- Anxiety
- Substance abuse



N3-4 Restorative sleep

REM Dreaming

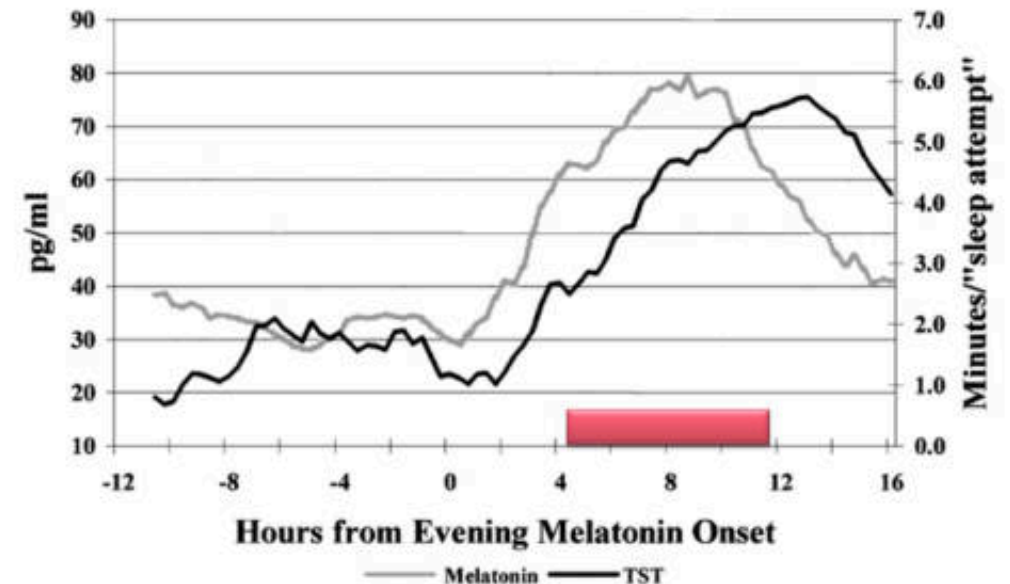
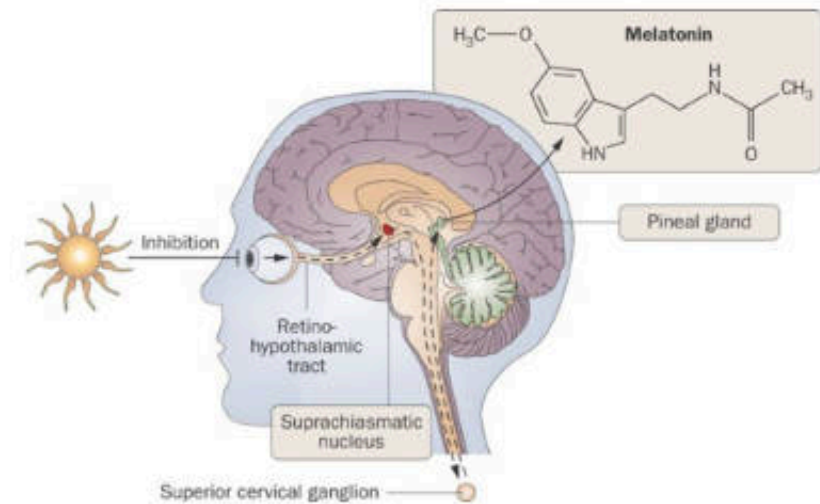
Learning and memory

Neurotransmitters involved in sleep-wake cycle

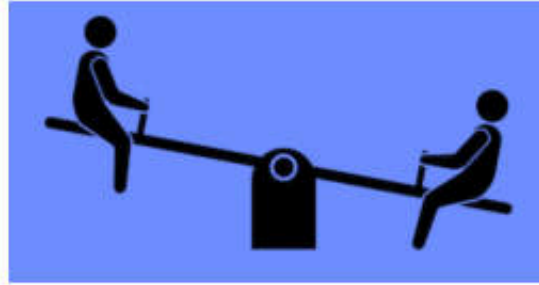
- ✧ Melatonin → opener of the sleep gate
 - ★ Released from pineal gland
 - ★ Trigger releasing of GABA to inhibit monoaminergic nuclei
- ✧ Monoamines
- ✧ Neuropeptide orexin

Melatonin

- ✱ A neurohormone released from pineal gland
- ✱ Essential for circadian rhythm
- ✱ Bind to MT1 and MT2
- ✱ Trigger GABA release from ventrolateral preoptic nucleus (VLPO) and the opening of the sleep gate
- ✱ *1.5–2 h lag between the onset of melatonin secretion*
- ✱ Peak level at 2-4 am
- ✱ Reduced level in elderly



Flip-flop switch model of sleep-wake control



Awake

Sleep

Inputs from sensory ctx, cortical/subcortical area, etc.

Inputs from sensory ctx, cortical/subcortical area, etc.

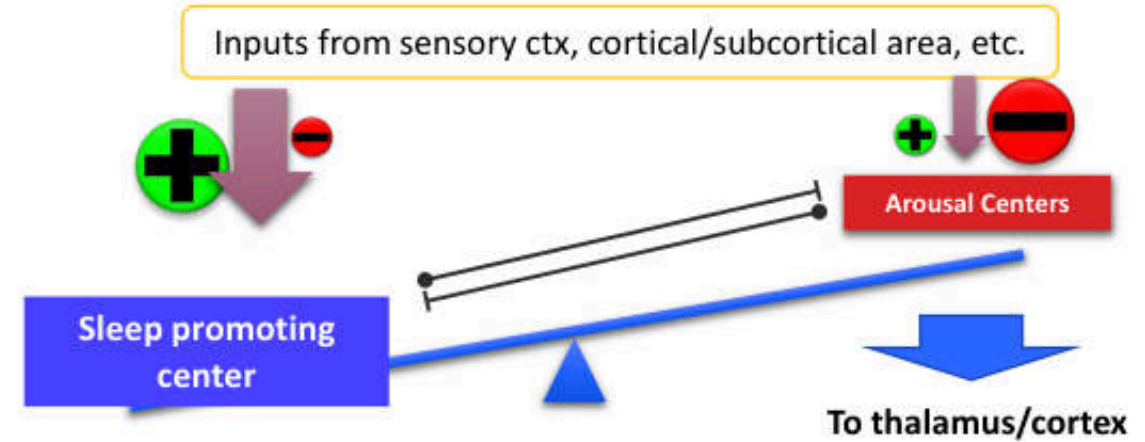
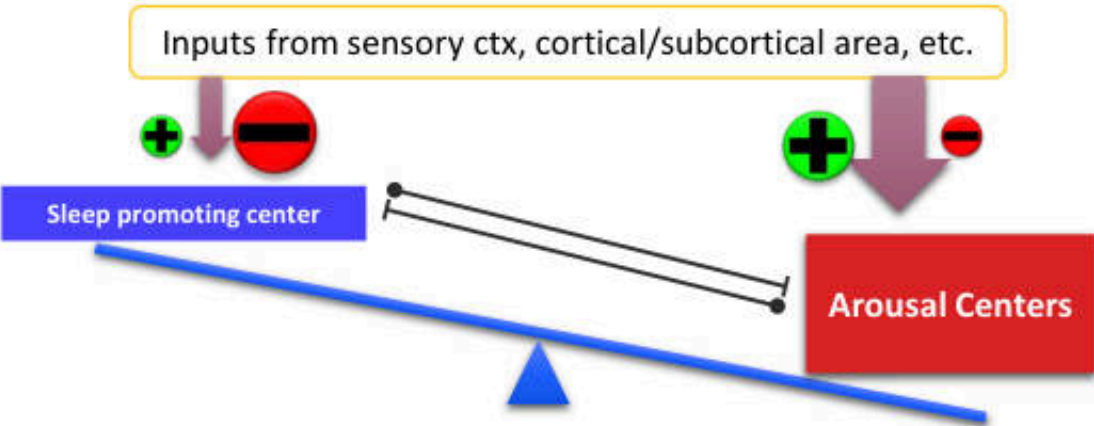
Sleep promoting center

Arousal Centers

Sleep promoting center

Arousal Centers

To thalamus/cortex



Inputs from sensory ctx, cortical/subcortical area, etc.

  
**Sleep promoting
center**

  
Arousal Centers

LH
(ORX)

RN
(5-HT)

LC
(NE)

TMN
(His)

VTA
(DA)

NBM
(ACh)

PPT/LDT
(ACh)

VLPO

**Ventrolateral
preoptic
nucleus
(GABA)**



Orexin receptor

GABA_A

To thalamus/cortex

Inputs from sensory ctx, cortical/subcortical area, etc.

  
**Sleep promoting
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Melatonin



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Orexin receptor

GABA_A

To thalamus/cortex



Inputs from sensory ctx, cortical/subcortical area, etc.

  
**Sleep promoting
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Arousal Centers

Melatonin

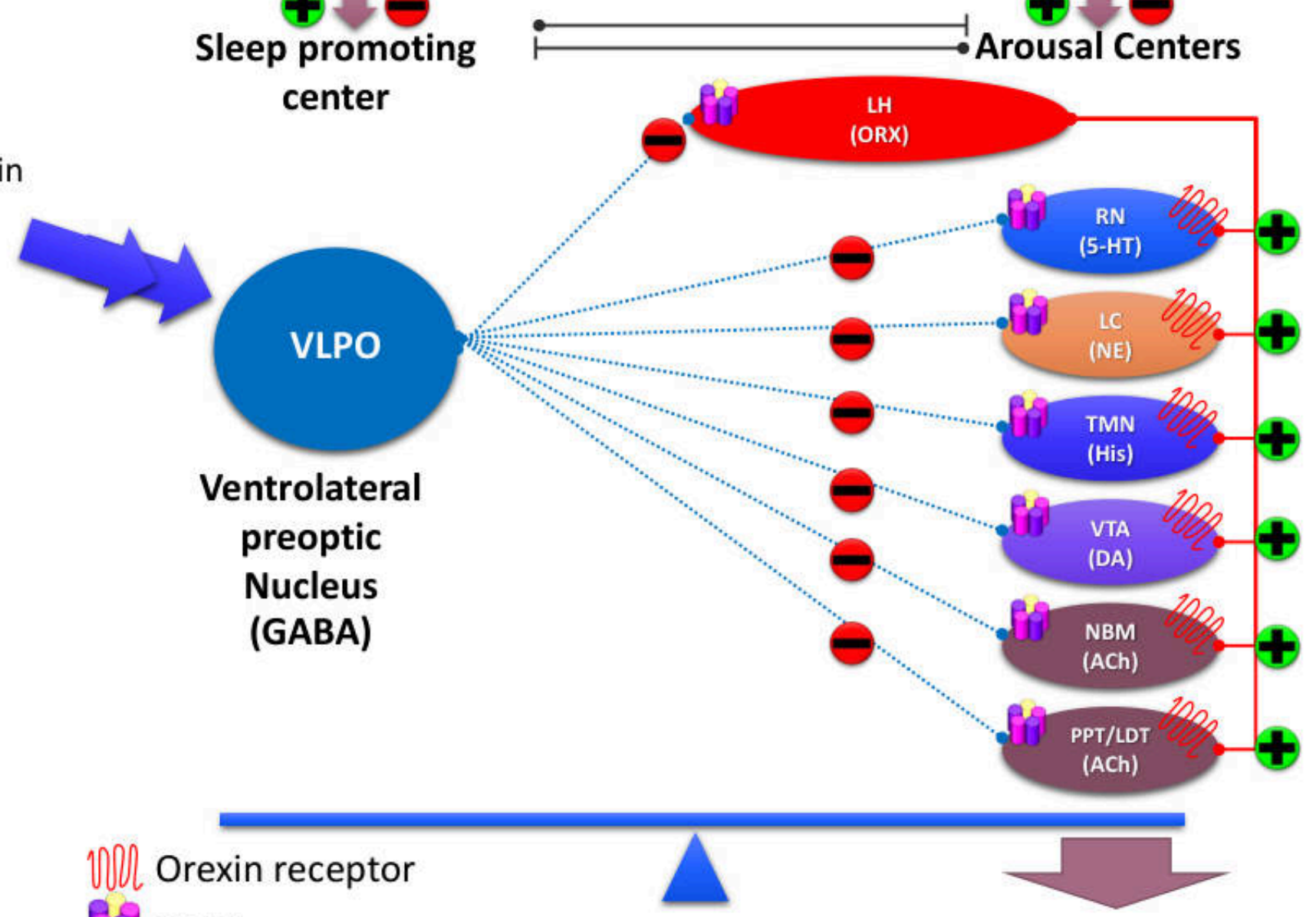


**Ventrolateral
preoptic
Nucleus
(GABA)**



 Orexin receptor
 GABA_A

To thalamus/cortex



Inputs from sensory ctx, cortical/subcortical area, etc.

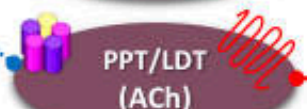
  
**Sleep promoting
center**

  
Arousal Centers

Melatonin



**Ventrolateral
preoptic
Nucleus
(GABA)**



Orexin receptor

GABA_A



To thalamus/cortex



Firing pattern of monoaminergic nuclei in wakefulness, NREM and REM sleep

Nucleus	Neurotransmitter	Firing activity			Receptor
		Wakefulness	NREM	REM	
LH	Orexin	++++	+	+	OX1, OX2
Raphe	5-HT	++++	++	0	Awake: various REM-off: 5-HT _{2A/2C}
LC	NE	++++	++	0	Awake: various REM-off: α_1 , α_2
TMN	Histamine	++++	++	0	H ₁
VTA	DA	++++	++	+	D ₁ , D ₂
NBM	ACh	++++	+	++++	M ₁
PPT/LDT	ACh	++++	0	++++	M ₁

Sleep REM-off
neurons

Sleep REM-on
neurons

LH=lateral hypothalamus, Raphe=raphe nuclei, LC=locus coeruleus, TMN = tuberomamillary nucleus, VTA=ventral tegmental area, NBM=nucleus basalis of Meynert, PPT = pedunculopontine nucleus, LDT=laterodorsal tegmental nucleus

Hypnotic drugs

1. Benzodiazepines
2. Z-drugs
 - ★ Zolpidem (Stilnox[®])
3. Melatonin and analogues
 - ★ Melatonin (Circadin[®])
 - ★ Ramelteon (Rozerem[®])
4. Orexin receptor antagonists
 - ★ Lemborexant (Dayvigo[®])
 - ★ Suvorexant (Belsomra[®])
 - ★ Daridorexant (Quviviq[®])
5. Chloral hydrate
7. Sedating antihistamines
 - ★ Hydroxyzine
 - ★ Diphenhydramine
8. Sedating antidepressants (low dose);
 - ★ Amitriptyline, doxepin
 - ★ Trazodone, mirtazapine, agomelatine
9. Sedating antipsychotics (low dose);
 - ★ Aripiprazole, olanzapine

Melatonin



PK

- ★ Short T_{1/2} 1-2 hr
- ★ Bioavailability is poor and varied



Indication

- ★ Prolonged release -> Circadin[®] for primary insomnia in elderly



Less sedative effect and less associated with tolerance and withdrawal



Patient information

- ★ Take 1-2 hr before desired time to sleep
- ★ Use at a regular time
- ★ May take days to weeks to adjust circadian rhythm

Orexin receptor antagonists

- ✱ Orexin binds to OX1/OX2 and promotes activation of most monoaminergic nuclei → maintain awake
- ✱ Loss of orexin neurons causes narcolepsy
- ✱ Blockade of OX1/OX2 promotes sleep
 - ★ Dual orexin receptor antagonists (DORAs); lemborexant, suvorexant, daridorexant
- ✱ Patient advice for lemborexant
 - ★ Take and go to bed immediately
 - ★ Have at least 7 hr sleep time

Advantages of DORAs vs benzodiazepines/Z-drugs

- ✱ Less N3 and REM suppression (may enhance)
- ✱ Less associated with tolerance and withdrawal
- ✱ Low risk of respiratory depression
- ✱ No motor side effects

General patient information

- ✳ Take at the appropriate time before desired time to sleep
- ✳ Avoid driving or operate heavy machinery
- ✳ Have sufficient night of sleep, at least 6-8 hr

Drugs with an USFDA indication for insomnia

Generic Name	Dose(s) (mg)	Metabolic Enzymes	Elimination Half-life (hours)	Tmax (hours)
Estazolam	1, 2	CYP3A4	10-24	0.5-6
Flurazepam	15, 30	CYP3A4 (probable)	47-100 ^a	0.5-1
Temazepam	7.5, 15, 22.5, 30	Conjugation with glucuronic acid	3.5-18.4	1.2-1.6
Triazolam	0.125, 0.25	CYP3A4	1.5-5.5	2
Quazepam	7.5, 15	CYP3A4 (minor: CYP2C9, CYP2C19)	39-73 ^a	2
Zolpidem	5, 10	CYP3A4, CYP2C9	1.4-4.5	1.6
Zolpidem CR	6.25, 12.5	CYP3A4, CYP2C9	2.8	1.5
Zaleplon	5, 10, 20	aldehyde oxidase (minor: CYP3A4)	1	1
Eszopiclone	1, 2, 3	CYP3A4, CYP2E1	6	1
Ramelteon	8	CYP1A2 (minor: CYP2C, CYP3A4)	1-2.6	0.75
Doxepin	3, 6	CYP2C19, CYP2D6 (minor: CYP1A2, CYP2C9)	15.3-31 ^b	3.5